

Procurement Projection & AI Planning

Executive Report — Fast&Up; (Aeronutrix Sports Products Pvt Ltd)

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Executive Summary

Fast&Up; has grown 3× in 3 years (Rs.3.5 Cr → Rs.10.5 Cr). This report presents a complete AI-driven procurement plan to achieve the next 3× growth target of **Rs.31.5 Cr in FY26** using a **Rs.10.08 Cr procurement budget** across 5 key SKUs. The plan integrates demand forecasting, intelligent reorder point management, linear programming budget optimization, and supplier risk assessment — delivering actionable recommendations for the procurement team.

Task 1: Demand Forecast

FY26 revenue of **Rs.31.24 Cr** was projected across 5 SKUs using historical growth trajectory analysis. Plant Protein Chocolate (3.8×) and Reload E&V; (3.5×) are the fastest-growing SKUs, while Reload Orange remains the volume leader at 2,10,000 units. Total forecasted units: **4,12,747**.

Task 2: AI Reorder Point Calculator

Reorder points were calculated using the safety stock formula: **ROP = (Daily Demand × Lead Time) + Safety Stock**, where Safety Stock = $Z \times \sigma_{LT} \times \sqrt{LT}$ ($Z=1.65$, 95% service level). Stress testing at 50% demand spike showed all SKUs require ~49% higher reorder points. Key finding: Plant Protein Chocolate carries highest procurement risk due to 30-day lead time and 75% supplier reliability score.

Task 3: Budget Optimization

Linear programming (PuLP) was used to optimally allocate the Rs.10.08 Cr budget. The optimizer maximizes revenue-weighted units while ensuring minimum 60% demand fulfillment per SKU.

SKU	Budget Allocated	Units Procured	Fulfillment %
Reload Orange	Rs.3.78 Cr	2,10,000	100.0%
Reload E&V	Rs.1.62 Cr	77,740	100.0%
Plant Protein Chocolate	Rs.2.63 Cr	36,480	71.6%
Lean Body	Rs.1.27 Cr	19,845	60.0%
Daily Fiber	Rs.0.79 Cr	24,574	60.0%
TOTAL	Rs.10.08 Cr	3,68,639	89.3%

Sensitivity Analysis: A 20% budget reduction to Rs.8.06 Cr reduces total procurement by 16.4% (307,292 units). A 20% budget increase to Rs.12.10 Cr adds 34,304 units (+9.4%), primarily benefitting Plant Protein Chocolate which is currently budget-constrained.

Task 4: Supplier Risk Assessment

15 suppliers assessed (3 per SKU) on 5 weighted dimensions. **3 suppliers flagged as high/moderate risk.** SupplierB-EV (China, 5.75/10) is the most critical — it is the sole China-dependent supplier for Reload E&V; with poor reliability and geo-risk scores. SupplierA-PP (Netherlands, 8.65/10) is the highest-scoring supplier overall — recommended as primary for Plant Protein.

Top 3 Procurement Risks & Mitigation

1. Plant Protein Budget Shortfall

Risk: 28.4% demand gap = Rs.2.61 Cr in lost revenue potential.

Mitigation: Request Rs.2 Cr budget top-up in Q2. Simultaneously negotiate 15% volume discount with SupplierA-PP to increase effective purchasing power.

2. China Supplier Concentration (Reload E&V;)

Risk: SupplierB-EV (5.75/10) is only China-based supplier for a fast-growing SKU.

Mitigation: Qualify South Asian alternative supplier within 90 days. Target composite score >7.5. Run parallel trial order to validate lead times.

3. Demand Spike Vulnerability

Risk: 50% demand spike increases all reorder points by ~49% — all SKUs could face stock-outs.

Mitigation: Build 30-day safety stock buffer for Reload Orange and Plant Protein ahead of summer and festival seasons. Automate reorder alerts via Python scripts.

90-Day Action Plan

Timeline	Action Items
May 2026	Place 26 POs for all 5 SKUs at optimized quantities · Begin SupplierB-EV replacement qualification · Deploy Python reorder a
June 2026	Review Plant Protein fulfillment + request Q2 budget top-up · Negotiate bulk discount with SupplierA-PP · Build Reload Orange safety s
July 2026	Qualify Reload EV backup supplier (South Asia) · Mid-year demand forecast review and adjustment · Supplier performance review — sco

AI/Automation Roadmap

- **Automated reorder alerts** (Immediate) — Python script triggering procurement team notifications when inventory crosses reorder point. Estimated setup: 2 days.
- **Demand forecasting pipeline** (Month 1) — Scheduled monthly reforecast using rolling actuals. Replaces manual Excel-based estimates.
- **Supplier performance dashboard** (Month 2) — Live Power BI dashboard scoring all 15 suppliers on actual delivery and quality data. Auto-flags underperformers.
- **Budget optimization engine** (Month 3) — Interactive tool allowing procurement team to run what-if budget scenarios before committing POs.

Tools: Python (pandas, numpy, matplotlib, PuLP). AI tools (Claude, Anthropic) used for code structuring and report drafting. All assumptions, analytical decisions, and business interpretations were reviewed by the author.